$$f(x) = \frac{3}{1 - x^2}$$

Find a power series for f.

Choose 1 answer:

INCORRECT

$3 + 3x + \ \ldots + \ 3x^n + \ldots$

INCORRECT

$3x + 3x^2 + \ldots + 3x^{n+1} + \ldots$

CORRECT (SELECTED)

$3 \ + \ 3x^2 \ldots + \ 3x^{2n} + \ldots$

INCORRECT

$3 \ + \ 9x^2 + \ldots + \ \left(3x^2\right)^n + \ldots$

1 / 2 This is a geometric series with first term $a \ = \ 3$ and common ratio $r \ = \ x^2$.

2 / 2 **Therefore, the series is as follows:**

$$3 \ + \ 3x^2 + \ 3x^4 + \ 3x^6 + \ldots + \ 3(x^2)^n + \ldots$$

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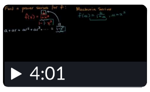

Function as a geometric series

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